

Prakhar Varshney

Full Stack Engineer @ Sourcy | CS @ BU | Building the Future of AI-Driven Commerce

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Summary

Accomplished Full Stack Engineer with a Master's in Computer Science, specializing in cutting-edge AI-powered solutions and scalable cloud architectures. Expert in designing and deploying advanced agentic applications, leveraging multi-agent architectures and progressive disclosure for enhanced automation and decision-making. Proficient in harnessing AWS, Python, JavaScript, and modern frameworks to develop and deploy complex systems, consistently optimizing performance and delivering high-impact products.

Experience

Full Stack Engineer

Aug 2024 – Present

Sourcy – Singapore

- Architected and deployed a state-of-the-art multi-agent Sourcing Agent, harnessing AWS (Lambda, CDK), Gemini, OpenAI, and Anthropic models to automate end-to-end specification generation and product shortlisting, processing over 1200 monthly requests with 95% accuracy.
- Optimized platform infrastructure to process over 100 SR specifications per minute, achieving a 70% reduction in turnaround time through serverless parallelization (AWS Lambda, Step Functions, SQS) and intelligent model orchestration.
- Spearheaded the development of critical sourcing automation features, including AutoQuotation generation and dynamic supplier shortlisting, driving a 30% increase in sourcing throughput leveraging React, Next.js, TypeScript, Python, PostgreSQL, and Zod.

Senior Software Engineer

Jan 2023 – Aug 2024

Patient First.AI – Boston, Massachusetts, United States

- Led the full engineering lifecycle for VacTrack, an AI-powered vaccination compliance platform, from conception to successful launch.

Software Engineer

May 2022 – Dec 2022

Patient First.AI – Boston, Massachusetts, United States

- Guided the design, development, and deployment of the Electronic Medical Record (EMR) system for physicians and patients.

Research Assistant

Dec 2021 – Sep 2022

Questrom School of Business, Boston University – Boston, Massachusetts, United States

- Collaborated on the INNOVAE project with Prof. DK Lee, converting unstructured patent data into actionable formats for research.

Graduate Teaching Assistant

Jan 2022 – May 2022

Boston University – Boston, Massachusetts, United States

Software Engineer

Jan 2021 – Dec 2021

TutorBin – Gurugram, Haryana, India

- Engineered and deployed expert and admin portals, enhancing marketing team throughput by 30% through Python, Django, Vue JS, Kubernetes, Docker, and AWS S3.
- Developed a microservice featuring an Event Sourcing mechanism for client-side payments via Kafka, accelerating payment processing by 65%.
- Implemented task-queue and invoice management with Django-Channels, Celery, and pyPDF, boosting platform speed by 18% and reducing server load and cost by 21%.

Education

Boston University

Aug 2021 – Jan 2023

Master of Science - MS, Computer Science

Earned a Master of Science in Computer Science with a 3.81 GPA. Led the Boston University Cyber Security Team (Terrier Bytes), securing 3rd place at the NCAE Cyber Games East Conference in 2022. Relevant coursework included Analysis of Algorithms, Operating Systems, Data Science with Python, Web Analytics and Data Mining, Computer Networks, Artificial Intelligence, Secure Software Development, Directed Study (IoT and Robotics), Foundations of Machine Learning, and Computer Language Theory.

Skills

Programming Languages: Python, JavaScript, TypeScript, Java, Dart, C++, GDScript, HTML, CSS

AI/ML & Agentic Systems: Artificial Intelligence, Machine Learning, Generative AI, Agentic AI Applications, Multi-Agent Architecture, Emotion Detection, Face Recognition, Time Series Analysis, MLOps

Web Frameworks & Libraries: React.js, Next.js, Vue.js, Django, Express.js, Node.js, Flutter, Socket.io, Zod

Cloud & DevOps: AWS (Lambda, CDK, S3, EC2, Step Functions, SQS), Kubernetes, Docker, CI/CD, Serverless Architecture

Databases: PostgreSQL, MongoDB

Tools & Technologies: Kafka, Celery, pyPDF, Arduino, CLI, Git, Jupyter Notebook, Selenium, REST APIs, Microservices, Data Mining, Algorithms

Open Source

- **smart-irrigation-system-Python** (8 ★)
Engineered an Arduino-based smart irrigation system that transmits live data to a connected ID, enabling users to precisely configure moisture levels and select specific crops.
- **Meet.OZ** (1 ★)
Created Meet.OZ, an AI-driven platform designed for organizations and educational institutions to analyze participant emotions during online meetings and classes.
- **Wordle-AI** (1 ★)
Developed Wordle-AI, an agent leveraging heuristic and non-heuristic search techniques to efficiently determine correct letters and sequences for solving Wordle puzzles.
- **api-cli** (2 ★)
Constructed a CLI tool to automate boilerplate code generation and dependency installation, streamlining rapid REST API setup with Express, body-parser, and MongoDB.
- **Holistic** (4 ★)
Designed Holistic, a mobile application focused on helping users track and manage their mental well-being.